Key answer for short answer question

8. Concentration of HNO_3 = 0.04M

$[H_3O^+]=0.04 \text{ mol dm}^{-3}$

$pH=-\log[H_3O^+]$

$=-\log(0.04)$

$=-\log(4\times10^{-2})$

$=2-\log4$

$=2-0.6021$

$=1.3979 = 1.40$

14. $Ba(OH)_2 \rightarrow Ba^{2+}+2OH^-$

$1.5\times10^{-3}M \qquad 2\times1.5\times10^{-3}M.$

$[OH^-] = 3\times10^{-3}M$

$[\because pH+pOH=14]$

$pH= 14-pOH$

$pH= 14- \left(-\log [OH^-]\right)$

$= 14+ \log [OH^-]$

$=14+\log(3\times10^{-3})$

$=14 + \log 3 + \log 10^{-3}$

$= 14+ 0.4771-3$

$=11+0.4771$

$pH=11.48$

15. Number of moles of $HNO_3 = 0.05 \times 50 \times 10^{-3}$

$=2.5 \times 10^{-3}$

Number of moles of $KOH = 0.025 \times 50 \times 10^{-3}$

$= 1.25 \times 10^{-3}$

Number of moles of HNO_3 after mixing

$= 2.5 \times 10^{-3} - 1.5 \times 10^{-3}$

$= 1.25 \times 10^{-3}$

$\therefore$ concentration of $HNO_3 = \dfrac{\text{Number of moles of } HNO_3}{\text{Volume is litre}}$

After mixing, total volume = 100 ml = 100×10^{-3}L

$\therefore [H^+]=\dfrac{1.25\times10^{-3}\text{moles}}{100\times10^{-3}\text{L}}$

$=1.25\times10^{-2}$ moles L^{-1}

$pH = -\log [H^+]$

$pH = -\log(1.25\times10^{-2}) = 2-0.0969$

$= 1.9031$

16. Given

$K_a=10^{-9}$

$c=0.4M$

$pH=-\log[H^+]$

$[H^+]=\sqrt{K_a\times c.}$

$=\sqrt{10^{-9}\times0.4}$

$=2\times10^{-5}$

$\therefore pH= -\log(2\times10^{-5})$

$=5-\log2$

$=5-0.3010$

$=4.699.$

17. $h = \sqrt{K_h} = \sqrt{\dfrac{K_W}{K_aK_b}} = \sqrt{\dfrac{1\times10^{-14}}{1.8\times10^{-5}\times1.8\times10^{-5}}}$

$= \dfrac{1}{1.8}\times10^{-2}$

$= 5.556\times10^{-3}$

$pH = \frac{1}{2} pK_w + \frac{1}{2} pK_a - \frac{1}{2} pK_b$

Given that $K_a = K_b = 1.8 \times 10^{-5}$

if $K_a = K_b$, then, $pK_a = pK_b$

$\therefore pH = \frac{1}{2} pK_w = \frac{1}{2} (14) = 7$

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